# DESEÑO E CONSTRUCIÓN DUN PROTOTIPO PARA O ESTUDO DO CONSUMO ENERXÉTICO EN VIVENDAS

POR SAÚL CID SEIJAS

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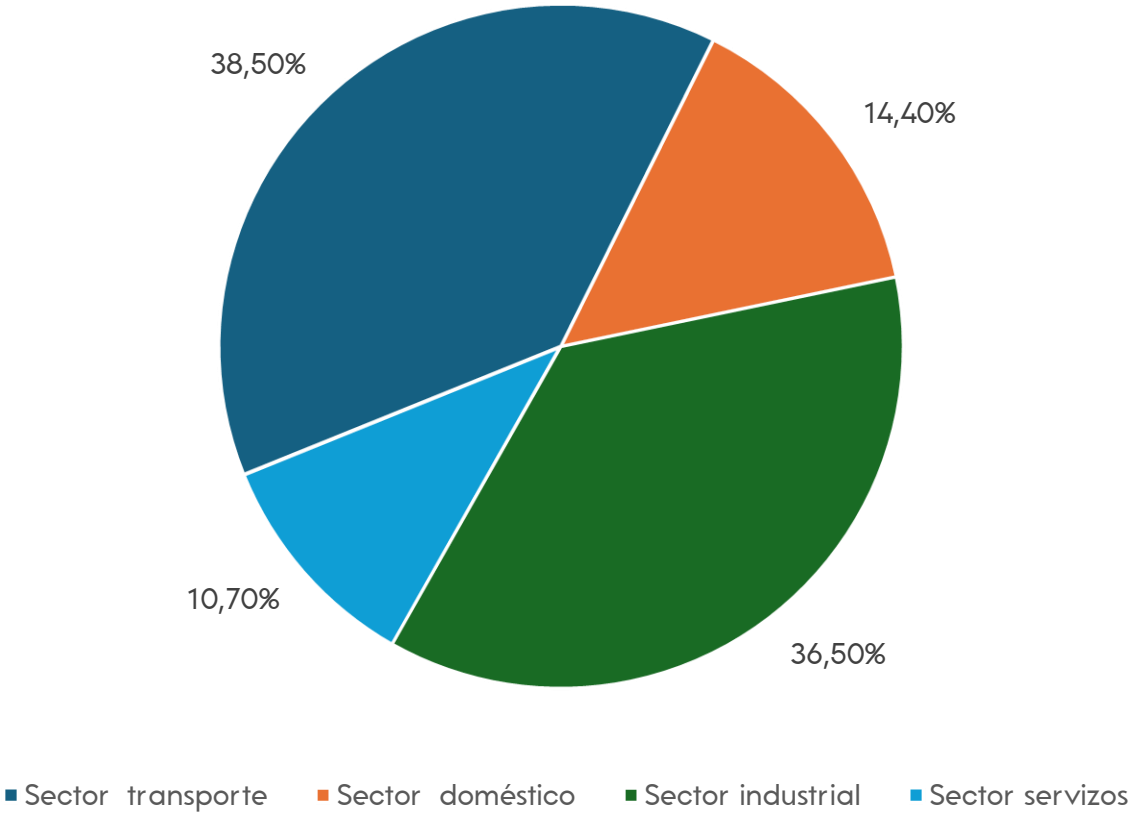
TITOR USC: XOSE RAMÓN FERNÁNDEZ

TITORA IES: VERÓNICA SANTISO VARELA

# LIMIAR

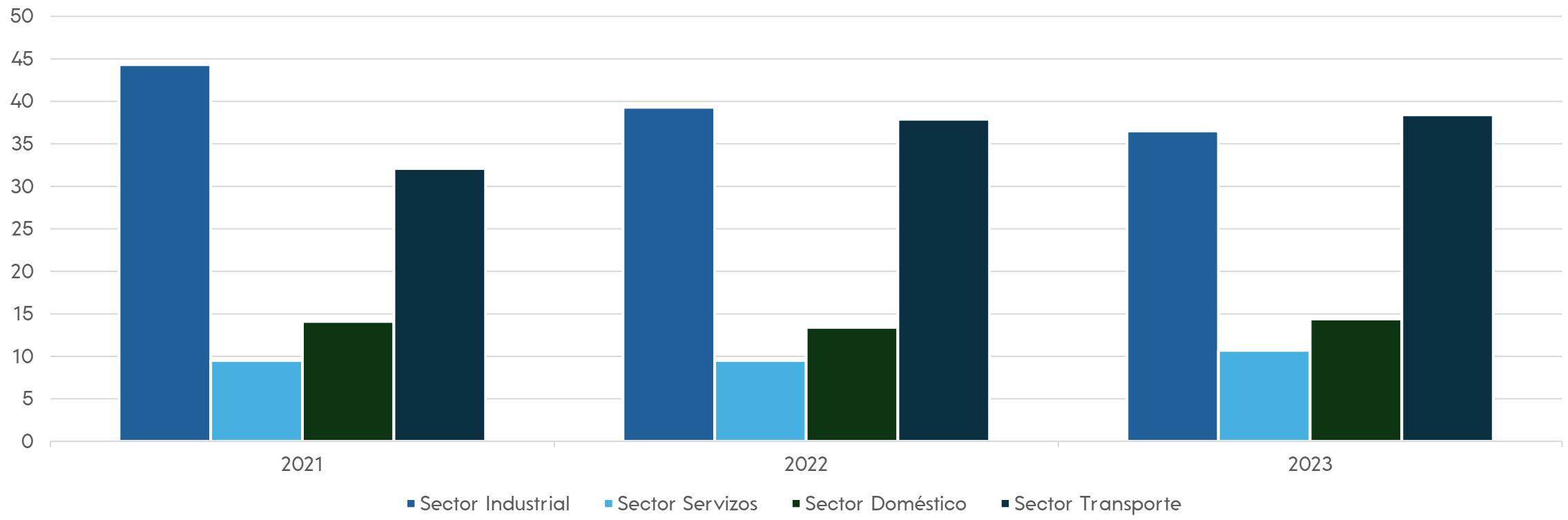
## A CRISE CLIMÁTICA

# CONSUMO ENERXÉTICO EN GALIZA




Datos: INEGA

# DISTRIBUCIÓN DO CONSUMO GALEGO POR SECTORES




Datos: IDAE

# OBXECTIVOS

- Estudo do illamento térmico
- Deseño da maqueta
- Deseño do sistema informático

# ESTADO DO COÑECEMENTO

# TRANSFERENCIA DE CALOR

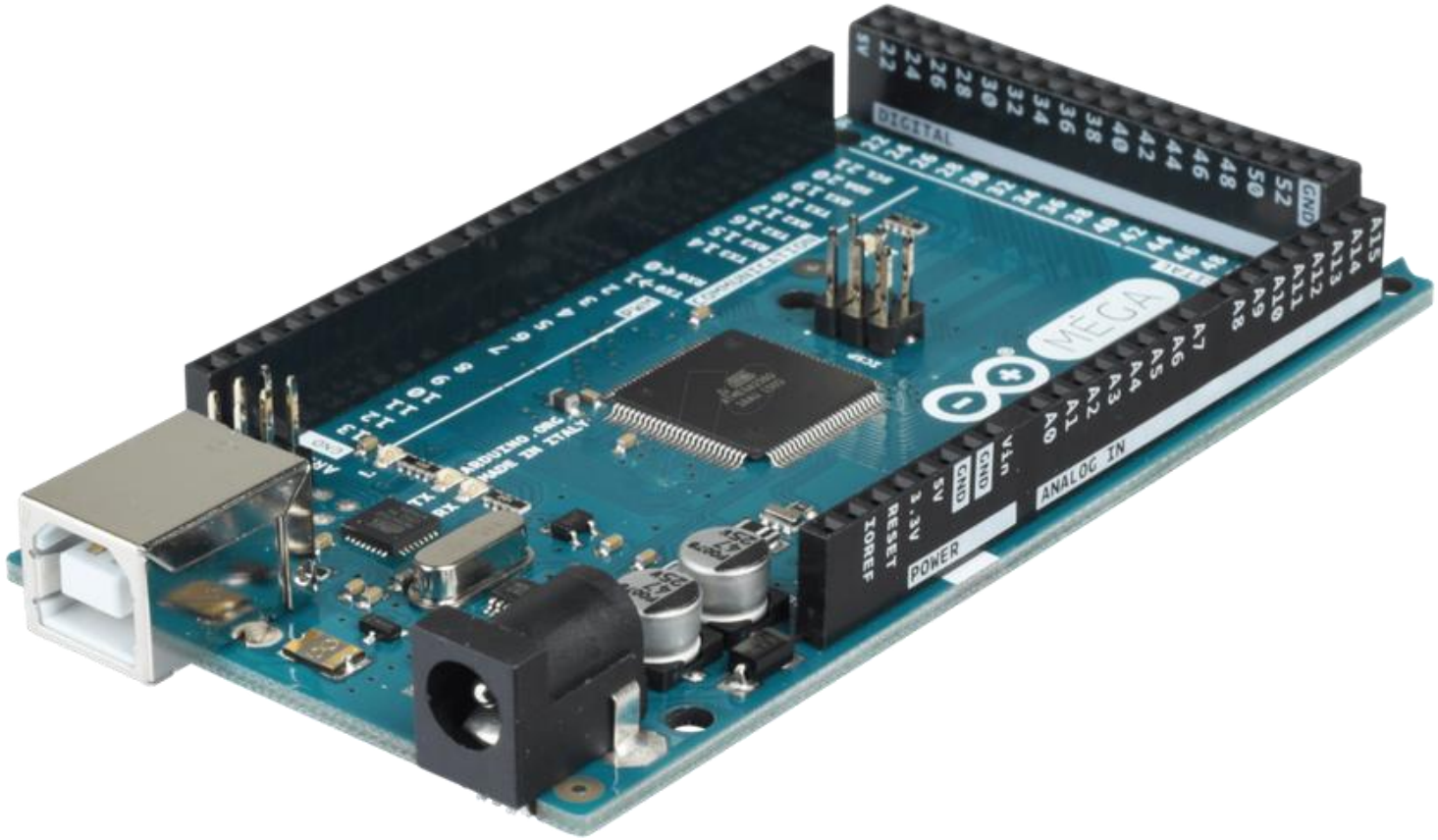
Convección

Radiación

Conducción

# ARDUINO

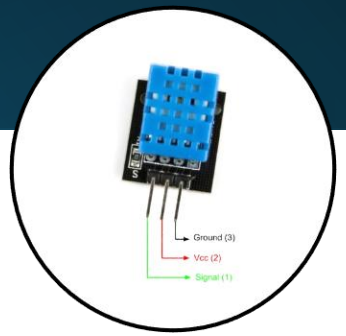
## Arduino Mega 2560



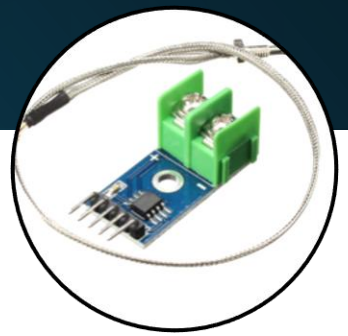
# SENSORES

| Sensor | Magnitude | Rango de medida | Precisión |
|---|---|---|---|
| **DHT11** (OSEPP Electronics) | Temperatura e humidade | 0 ºC a 50 ºC<br>20–90 %RH | ±2 ºC<br>±5 %RH |
| **DHT22** | Temperatura e humidade | -40 ºC a 80 ºC<br>0–100 %RH | ±0,5 ºC<br>±2–5 %RH |
| **DS18B20** | Temperatura | -55 ºC a 125 ºC | ±0,5 ºC |
| **MAX6675** | Temperatura | -20 ºC a +85 ºC | ±2 °C |
| **Fotorresistencia** (LDR) | Intensidade luminosa | Depende da luz ambiental | Comportamento non lineal |

# SENSORES

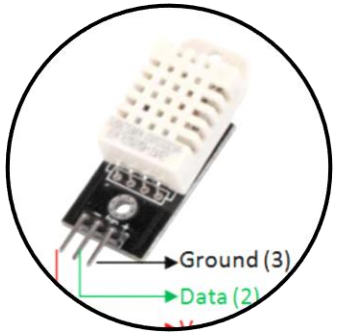



DHT11

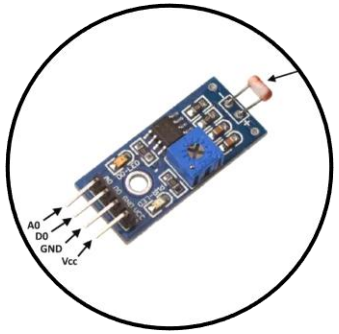


MAX6675

DS18B20



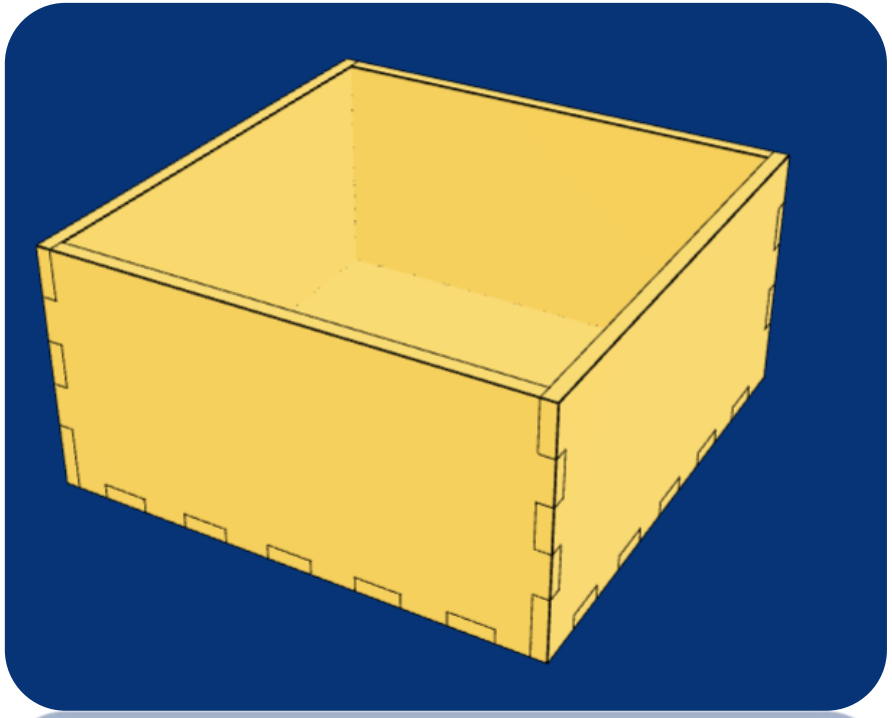

DHT22

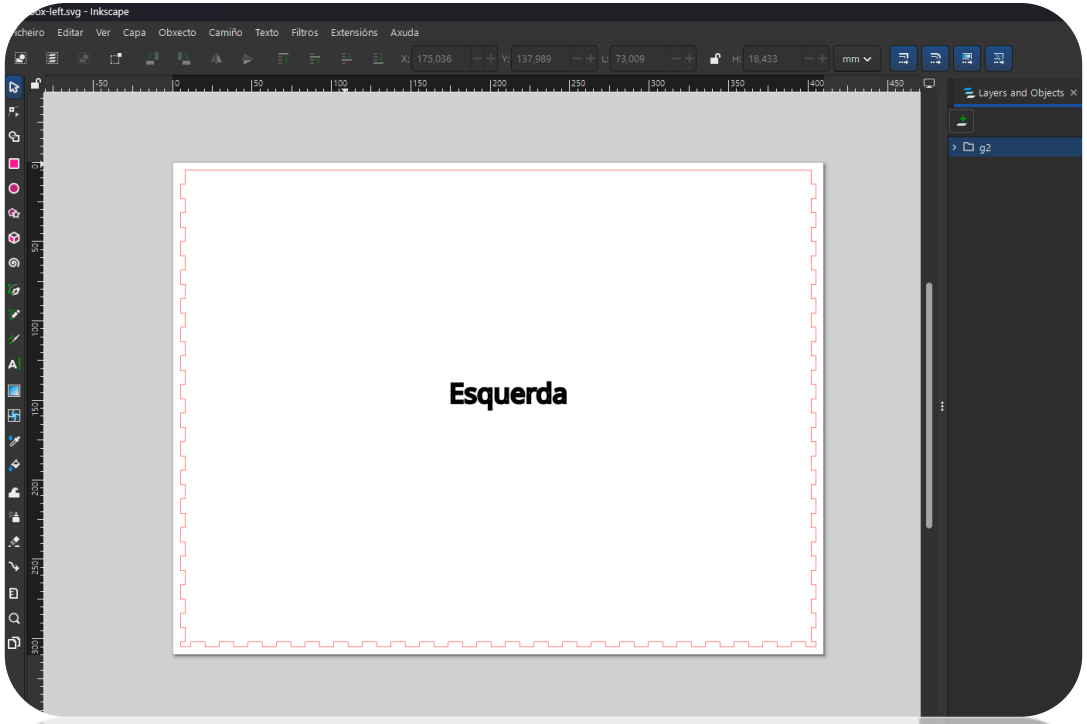



LDR

# O PROTOTIPO

# DESEÑO DA MAQUETA

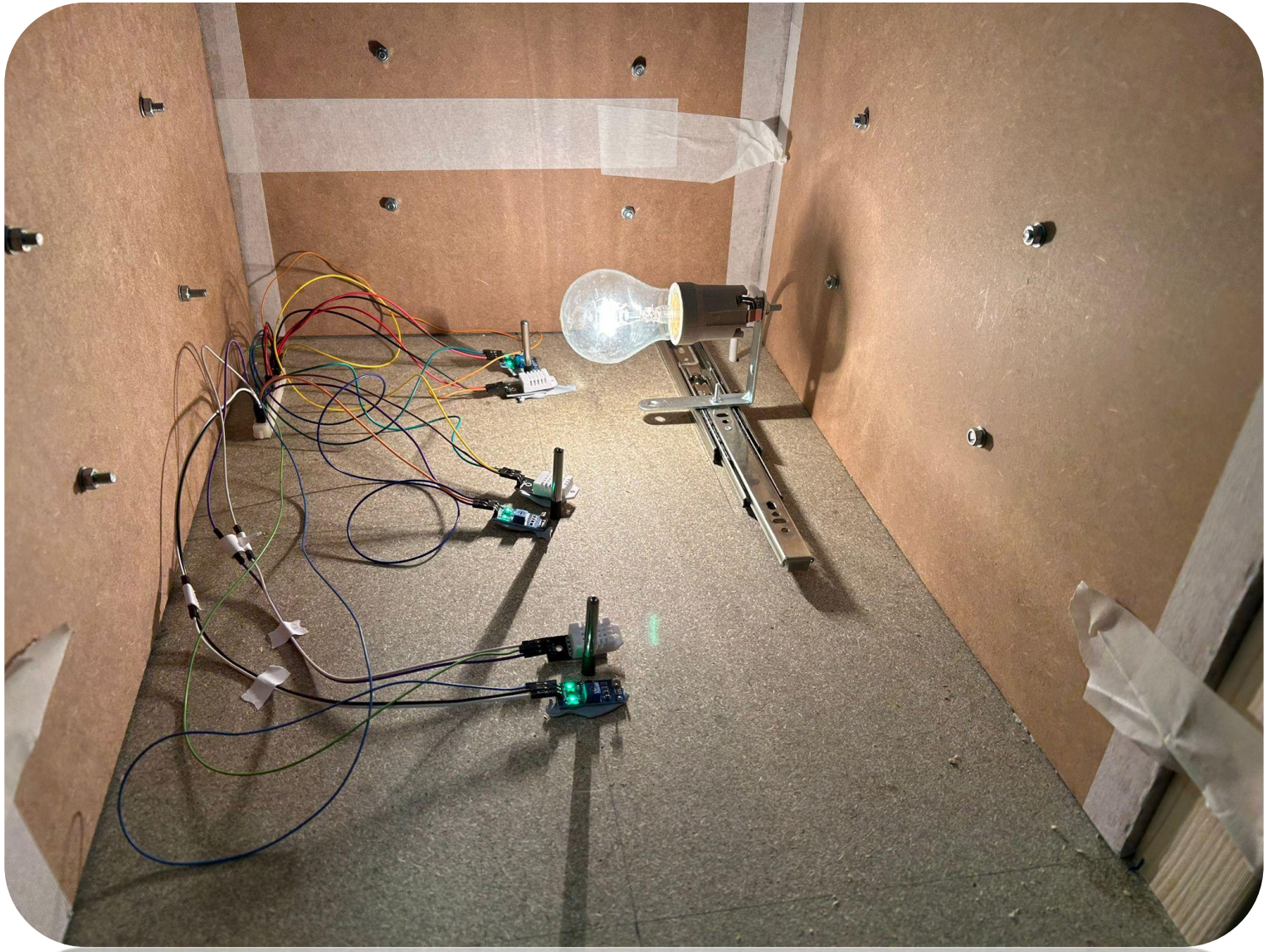




# CORTADORA LÁSER

# ESCUMA ILLANTE

# O PORTALÁMPADAS



# O ARDUINO

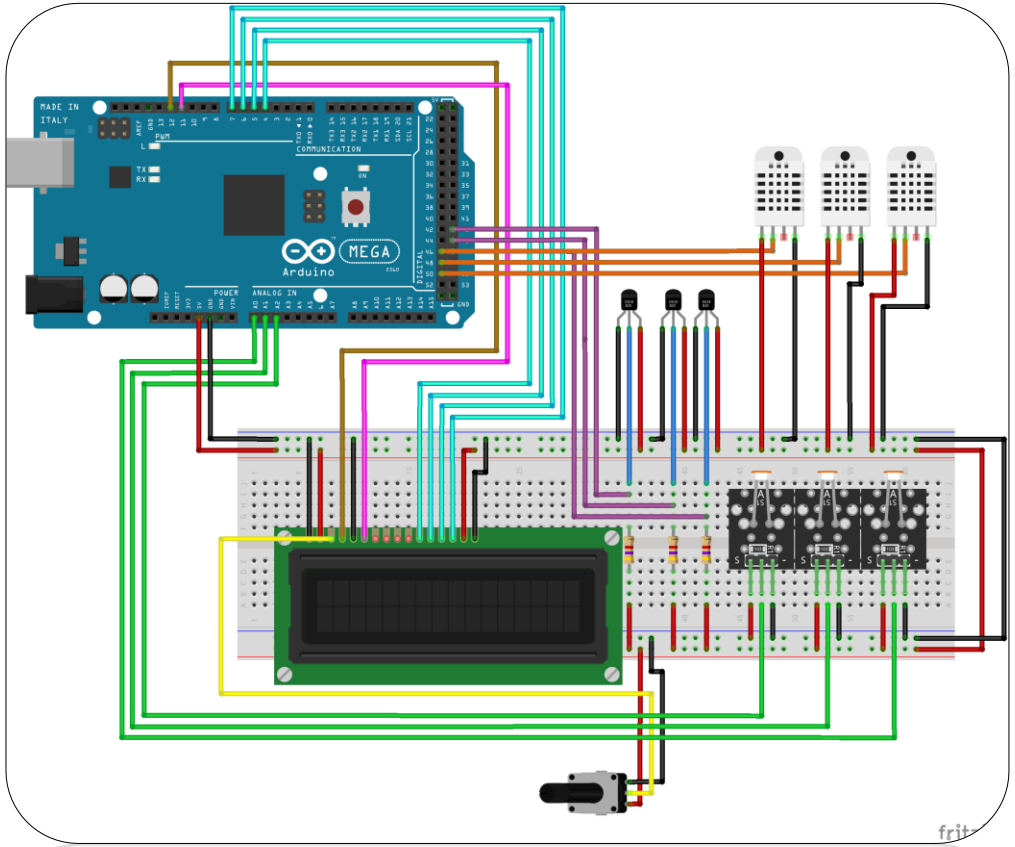
Proxecto | Arduino IDE 2.3.7

Archivo Editar Sketch Herramientas Ayuda

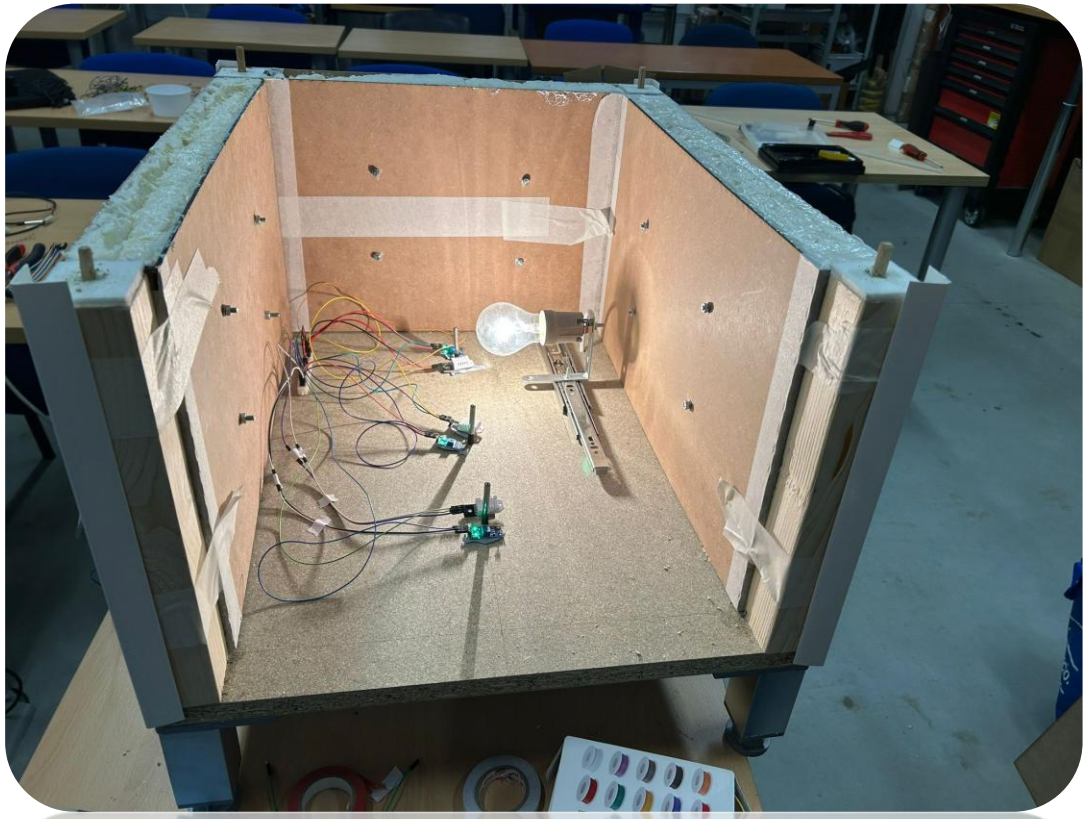
Arduino Mega or Mega 2...

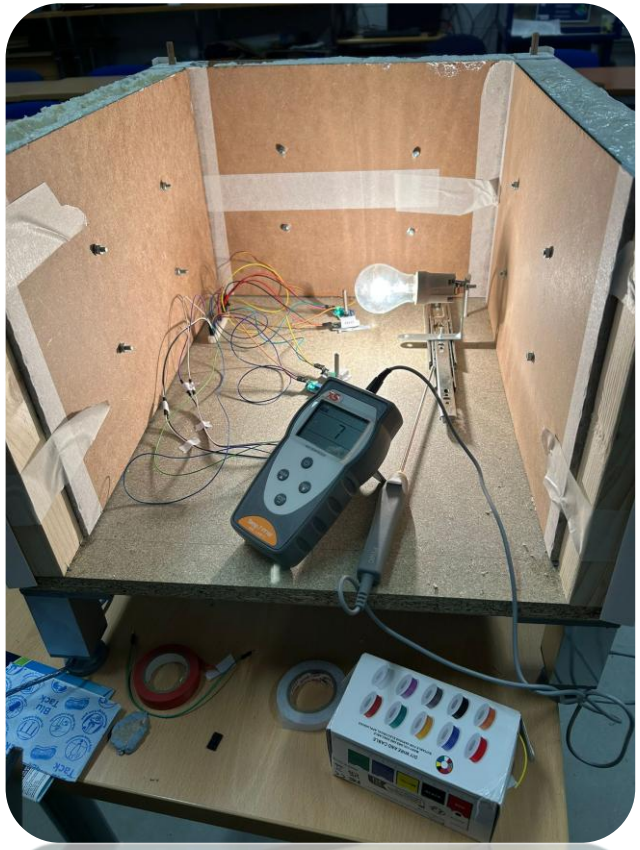
Proxecto.ino

```
#include <OneWire.h>
#include <DallasTemperature.h>
#include <LiquidCrystal.h>
#include <DHT.h>//libraria sensores dht
#include <SPI.h>//libraria sensores MAX6675 (Termopar)
//LCD
const int rs = 12, en = 11, d4 = 2, d5 = 3, d6 = 4, d7 = 5;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);

//Pin Botón
const int boton = 8;

//DHT22
const int DHT22PIN1 = 48;
const int DHT22PIN2 = 46;
const int DHT22PIN3 = 52;

DHT dht1(DHT22PIN1, DHT22);
DHT dht2(DHT22PIN2, DHT22);
DHT dht3(DHT22PIN3, DHT22);

float hum1,hum2,hum3,hum4,hum5;
float  temp1,temp2,temp3,temp4,temp5,tempMAX1,tempMAX2,tempMAX3,tempDS1,tempDS2,tempDS3;

//MAX6675 Termopar 1
//const int thermoDO1 = 49;
//const int thermoCS1 = 51;
//const int thermoCLK1= 53;

//Sensor DS18b20
const int pinDatos1DQ = 43;
const int pinDatos2DQ = 45;
const int pinDatos3DQ = 47;

OneWire oneWireObjeto1(pinDatos1DQ);
OneWire oneWireObjeto2(pinDatos2DQ);
OneWire oneWireObjeto3(pinDatos3DQ);
```



# RESULTADOS





# CONCLUSIÓN